





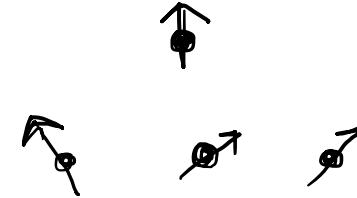
Noise Model (II)

Noise models: Modeling noise in Quantum

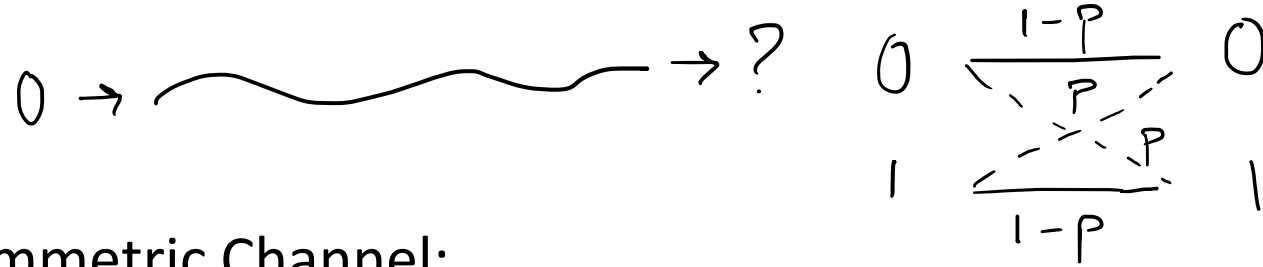
Pinakin Padalia

What are we covering in this chapter?

- Noise models: Noise and relevance for NISQ era
- **Noise models: Modeling noise in Quantum**
 - Errors in Classical and Repetition Codes
 - Errors in Quantum
 - Bit-flip Channel
 - Phase-flip Channel
 - Depolarizing Channel
 - Dephasing Channel
 - Amplitude Damping Channel



Errors in Classical and Repetition Codes



$$P_{e_1} = p$$

- Binary Symmetric Channel:

- Repetition Codes:

- Encoding by copying and majority voting

Simple!

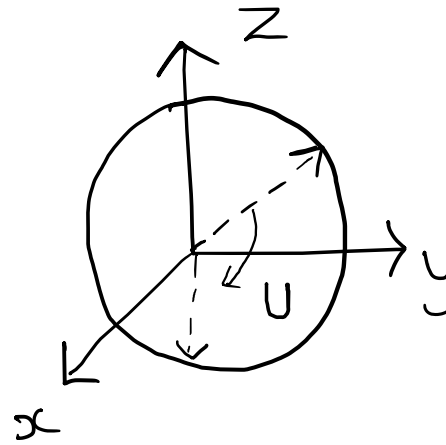
$0 \rightarrow 000 \rightarrow \text{~~~~~} \rightarrow ?$
 $1 \rightarrow 111$

$010 \equiv 000$
 $100 \equiv 000$
 $001 \equiv 000 \dots$

Cond.: $P_{e_2} < P_{e_1}$

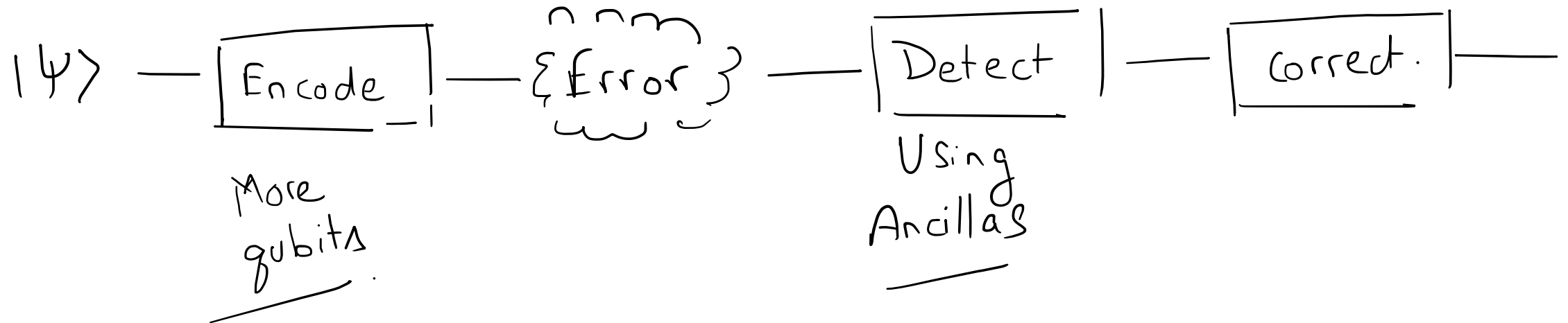
$\Rightarrow \underline{\underline{p < 1/2}}$ (verify!)

$$\begin{aligned}
 P_{e_2} &= p^3 + 3p^2(1-p) \\
 &= 3p^2 - 2p^3 \\
 &= p^2(3 - 2p)
 \end{aligned}$$



- Unitary operations & errors:
- Non-unitary operations & errors: 
- In general: errors take you from pure state to a mixed state
- Hurdles:
 - No cloning theorem — no copy allowed for ψ
 - Errors are continuous in nature — unlike classical.
 - Measurement destroys the state — so, no measure & restore

- Strategy (More about this in Module 5.2):



- Flipping of the bit: 0 $\rightarrow$ 1 and 1 $\rightarrow$ 0

- Quantum: Pauli X error = $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$|\psi\rangle \xrightarrow{P} X|\psi\rangle$$

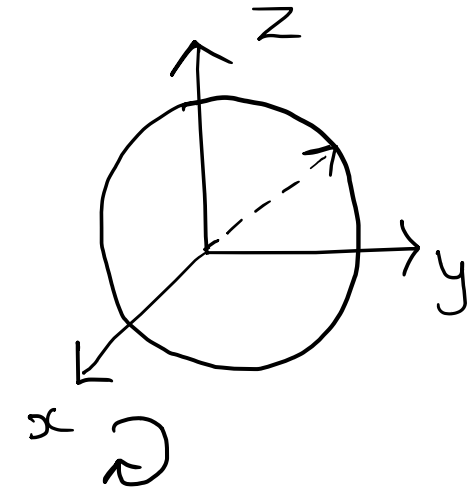
$$\swarrow 1-P$$

$$|\psi\rangle$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\downarrow P$$

$$X|\psi\rangle = \alpha|1\rangle + \beta|0\rangle$$



Phase Flip Channels

No classical eq.

- Phase flip errors: Pauli Z error = $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$

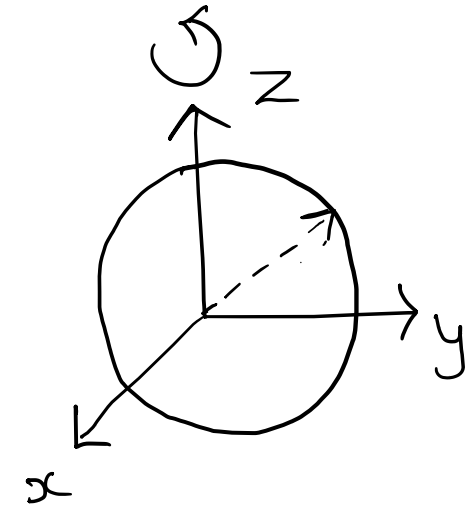
$$\begin{array}{l} |\psi\rangle \xrightarrow{p} Z|\psi\rangle \\ \quad \searrow^{1-p} \quad |\psi\rangle \end{array}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$\downarrow p$$

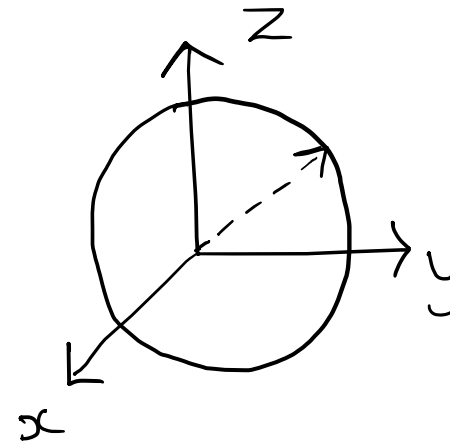
$$Z|\psi\rangle = \alpha|0\rangle - \beta|1\rangle$$

Phase flip
in $|0\rangle |1\rangle$
Basis $\equiv$ Bit flip
in $|+\rangle |-\rangle$
Basis

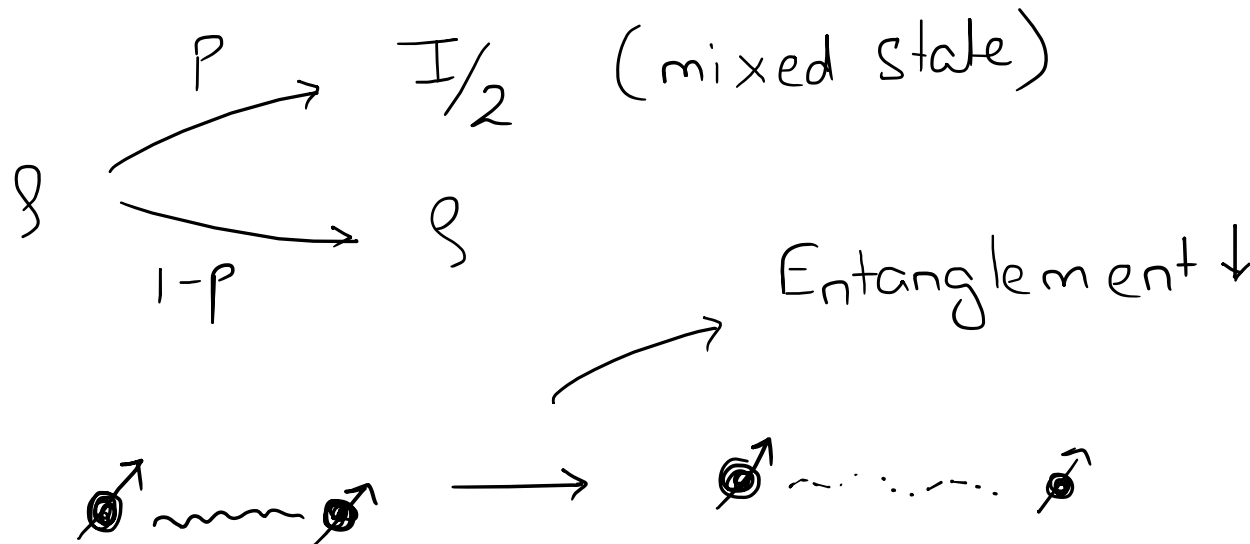
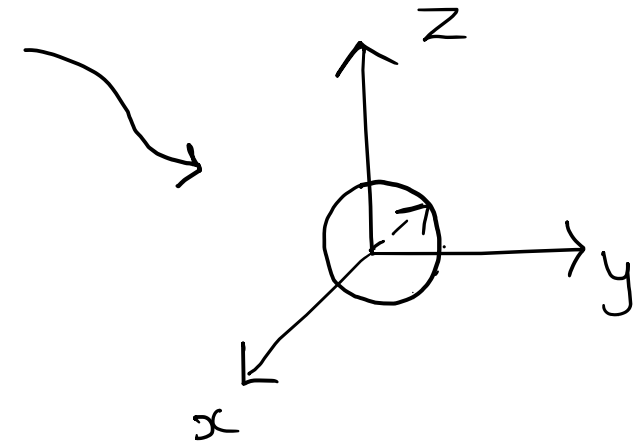


Depolarizing Channel

- Depolarizing noise:
 - Loss of information
 - The qubit loses superposition and goes to a mixed state
 - Shortens its length on Bloch sphere

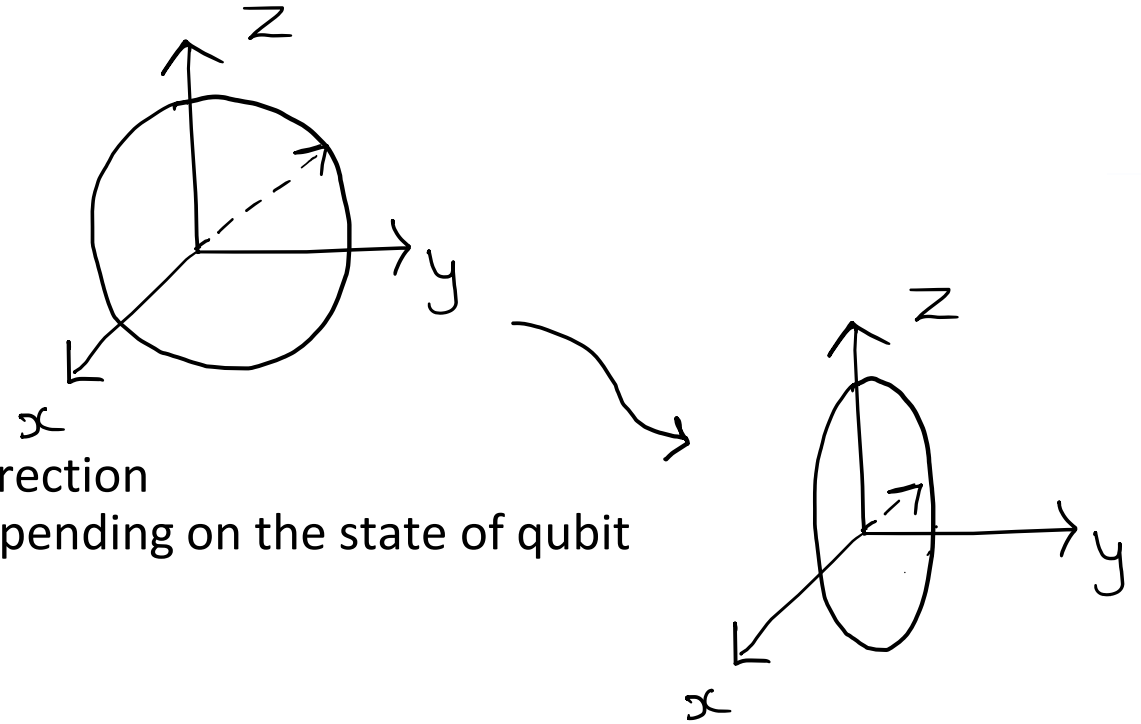


Interacts with env.

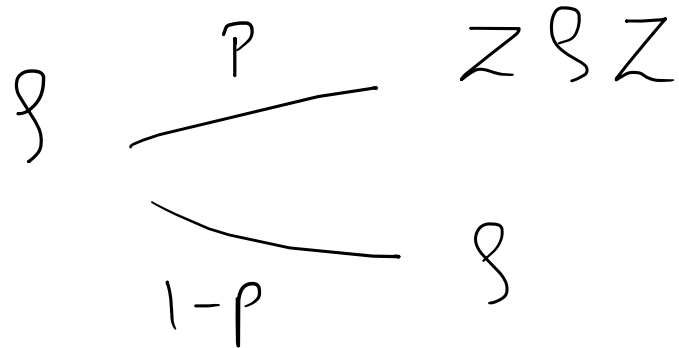


Dephasing Channel

- Dephasing noise:
 - Contraction of Bloch sphere in x and y direction
 - Environment "scattering" into a state depending on the state of qubit

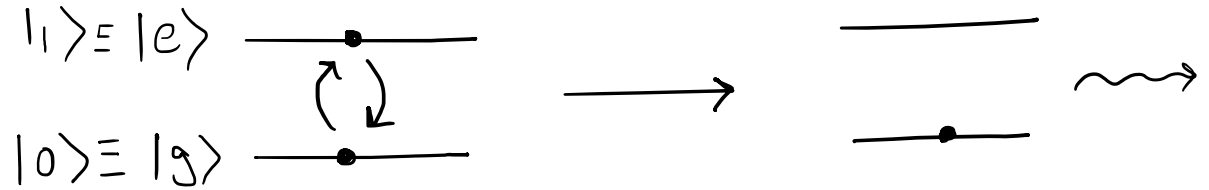


"Phase Damping Channel"



"measuring & forgetting"

Amplitude Damping Channel



- Amplitude Damping noise:
 - Common in photonic and spin systems
 - Decaying of excited state to ground state

$$|g\rangle \equiv |0\rangle$$

&

$$|e\rangle \equiv |1\rangle$$

$$\begin{array}{lcl}
 |0\rangle & \longrightarrow & |0\rangle \\
 |1\rangle & \xrightarrow{p} & |0\rangle \\
 & \searrow_{1-p} & |1\rangle
 \end{array}$$



Thank you for your attention!

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- R. Bavontaweepanya, "Effect of depolarizing noise on entangled photons", Journal of Physics: Conference Series, vol. 1144, p. 012047, 2018. Available: 10.1088/1742-6596/1144/1/012047.
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- J. Preskill, "Lecture Notes for Physics 229: Quantum Information and Computation", Lorentz.leidenuniv.nl, 1998. [Online]. Available: https://www.lorentz.leidenuniv.nl/quantumcomputers/literature/preskill_1_to_6.pdf.